RANK4 SIGNAL SYSTEMS

SPARC

Synchronized Phased Array RFSoC Cluster

A scalable digital beamforming architecture leveraging RFSoC 4x2 platforms, White Rabbit precision timing, and PPS synchronization for 48+ element coherent phased array systems.

CLASSIFICATION	PLATFORM	DISTRIBUTION
Company Confidential	RFSoC 4x2 / White Rabbit	Authorized Personnel
PREPARED BY	**TARGET SCALE**	**REFERENCE**
Rank4 Signal Systems	48+ Element Arrays	Project Phoenix / TSS

01 Executive Summary

Project SPARC (Synchronized Phased Array RFSoC Cluster) represents Rank4 Signal Systems' next-generation approach to scalable, coherent digital beamforming. Building upon the foundational concepts established in Project Phoenix for TSS Solutions, SPARC reimagines the architecture around the AMD/Xilinx RFSoC 4x2 development platform by RealDigital and introduces sub-nanosecond synchronization via White Rabbit precision timing protocol.

The core challenge in phased array systems at scale is maintaining coherent phase relationships across all antenna elements. Traditional approaches rely on centralized clock distribution, which becomes increasingly impractical beyond 16 elements due to cable length matching, thermal drift, and physical routing constraints. SPARC addresses this through a distributed timing architecture where each RFSoC node derives its sampling clock from a White Rabbit-disciplined reference, achieving sub-nanosecond synchronization across the entire array.

> Key Performance Targets: <1 ns inter-node synchronization accuracy | 48+ simultaneous antenna elements | <100 ps jitter on sampling clocks | Fully deterministic PPS distribution | Real-time adaptive beamforming across all elements

Each SPARC node is built around the RFSoC 4x2, which integrates high-speed ADCs and DACs directly alongside programmable logic fabric. This architecture eliminates the traditional bottleneck of routing RF signals through external digitizers, enabling each node to independently sample, process, and contribute to the composite beam pattern. The RFSoC 4x2 provides 4 ADC channels at up to 5 GSPS and 2 DAC channels at up to 9.85 GSPS, with integrated FPGA fabric for real-time signal conditioning and channelization.

White Rabbit, originally developed at CERN for particle accelerator timing, provides the synchronization backbone. By distributing a common frequency reference and a precise time-of-day stamp across a standard Ethernet infrastructure, White Rabbit achieves sub-nanosecond accuracy over fiber optic links spanning hundreds of meters. This is a fundamental enabler for SPARC, as it decouples the physical proximity requirement from the synchronization accuracy requirement.

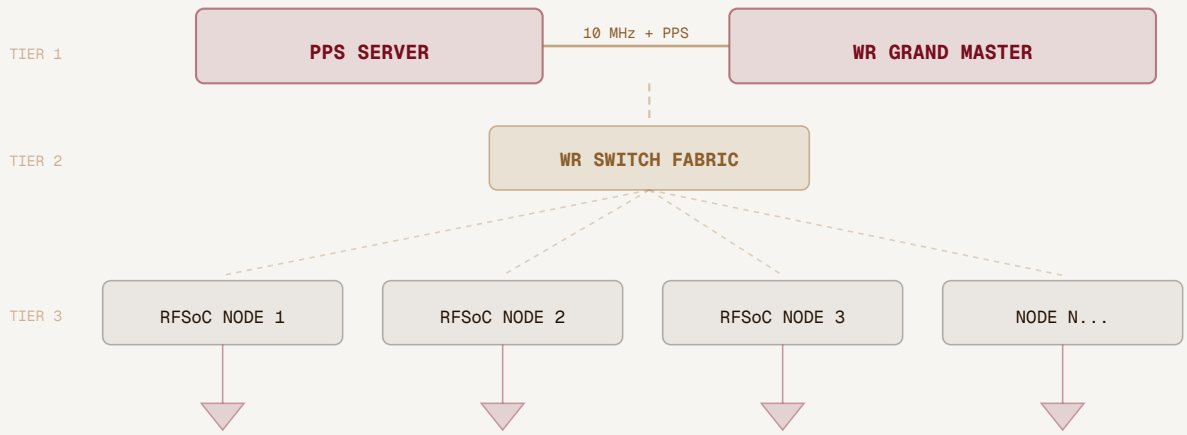
The PPS (Pulse Per Second) server completes the timing architecture by providing a universal epoch reference. Combined with White Rabbit's frequency lock, this enables deterministic timestamp alignment across all nodes, which is critical for coherent integration in post-processing and real-time beamforming alike. SPARC targets applications in electronic warfare, radar, communications, and scientific instrumentation where large-aperture, high-resolution phased arrays are required.

02 System Architecture

Distributed Timing & Processing Topology

The SPARC architecture is organized into three functional tiers: the Timing Distribution Layer, the Processing Node Layer, and the Coordination Layer. This separation of concerns ensures that timing integrity is maintained independently of processing load, while the coordination layer manages beamforming coefficients and array-level calibration.

FIG 1 // SPARC SYSTEM TOPOLOGY



Timing Distribution Layer (Tier 1): The PPS server generates a GPS-disciplined or atomic-referenced Pulse Per Second signal alongside a 10 MHz frequency reference. The White Rabbit Grand Master ingests these references and synthesizes a WR-disciplined clock that is distributed across standard single-mode fiber to all downstream nodes. This provides frequency lock, phase alignment, and absolute time-of-day to each node simultaneously.

Processing Node Layer (Tier 3): Each RFSoC 4x2 node receives the WR-disciplined clock and uses it to derive its local ADC/DAC sampling clocks. The on-chip PLL architecture of the RFSoC ensures low-jitter clock synthesis from the external reference. Each node independently digitizes antenna element signals, performs real-time channelization and filtering in FPGA fabric, and streams processed samples to the Coordination Layer via 10/25 GbE.

Coordination Layer: A host processing system aggregates channelized data from all nodes, applies beamforming weights (computed from calibration and steering vectors), and produces composite beam outputs. This layer also manages array-level calibration routines, including mutual coupling compensation and per-element phase/amplitude equalization.

03 Platform Components

Hardware & Synchronization Infrastructure

3.1 RFSoC 4x2 by RealDigital

The RFSoC 4x2 development platform provides a cost-effective, production-capable path to integrating high-speed data conversion with programmable logic. Built on the AMD/Xilinx Zynq UltraScale+ RFSoC family, it combines RF-class ADCs and DACs with a powerful FPGA and ARM processing subsystem on a single device.

RFSOC 4X2 KEY SPECIFICATIONS

ADC CHANNELS	4x 12-bit @ 5.0 GSPS
DAC CHANNELS	2x 14-bit @ 9.85 GSPS
FPGA FABRIC	ZU48DR UltraScale+ RFSoC
PROCESSING	ARM Cortex-A53 + R5F
CONNECTIVITY	10 GbE, QSFP28 (100G)
CLOCK INPUT	Ext. 10 MHz / PPS
POWER	12V DC, ~35W typical

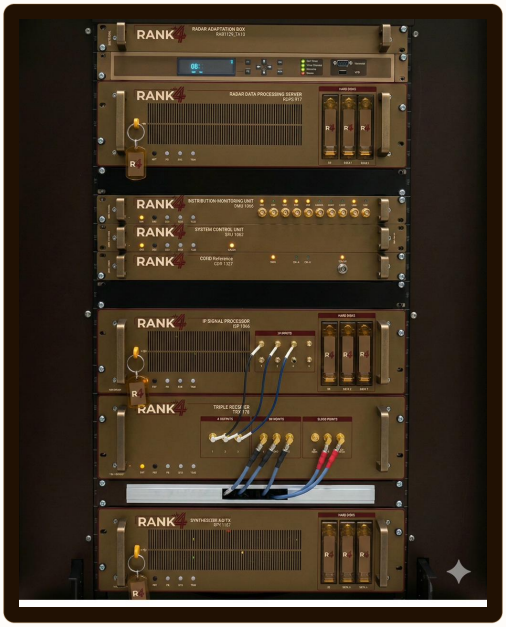


FIG 2 // RANK4 SIGNAL PROCESSING RACK
Proven heritage hardware platform

3.2 White Rabbit Synchronization

White Rabbit extends IEEE 1588 Precision Time Protocol (PTP) with synchronous Ethernet (SyncE) and a digital dual-mixer time-difference (DDMTD) phase detection scheme to achieve sub-nanosecond accuracy. Each WR node recovers both a frequency reference and an absolute time stamp from the Grand Master, enabling deterministic and repeatable clock alignment without manual cable calibration.

For SPARC, the WR switch fabric distributes timing to all RFSoC nodes over existing fiber infrastructure. The Grand Master accepts external 10 MHz and PPS from the PPS server and propagates a WR-locked reference to each endpoint. Each RFSoC node's WR endpoint extracts the recovered clock and feeds it to the RFSoC's external clock input, closing the synchronization loop with measured accuracy better than 100 ps RMS under typical conditions.

3.3 PPS Server & Epoch Reference

The PPS server anchors the entire timing hierarchy to an absolute time reference. Typically GPS-disciplined or driven by a rubidium/cesium oscillator, the PPS server outputs a clean 1 Hz pulse aligned to UTC along with a stable 10 MHz sinusoidal reference. This dual-output feeds the White Rabbit Grand Master, which in turn distributes coherent timing to the entire SPARC array. The PPS epoch alignment ensures that all nodes share a common notion of absolute time, enabling coherent integration windows and deterministic data tagging across the full array aperture.

04 Scalability & Beamforming

Expanding to 48+ Elements and Beyond

SPARC's architecture is fundamentally modular. Each RFSoC 4x2 node handles 4 antenna elements via its ADC channels. Scaling to a 48-element array requires 12 RFSoC nodes, all synchronized through the White Rabbit fabric. The architecture places no fundamental limit on array size — the WR switch fabric can cascade to support hundreds of endpoints, limited only by the fiber infrastructure and the coordination layer's processing bandwidth.

SCALING MATRIX

ELEMENTS	RFSOC NODES	WR ENDPOINTS	DATA RATE	BEAMS
16	4	4 + 1 GM	40 Gbps	4
48	12	12 + 1 GM	120 Gbps	12+
96	24	24 + 2 GM (cascade)	240 Gbps	24+
192+	48+	Cascaded WR fabric	480+ Gbps	48+

Beamforming Architecture: SPARC supports both time-domain and frequency-domain beamforming. In time-domain mode, true time delays are applied per-element in FPGA fabric to maintain broadband coherence. In frequency-domain mode, per-channel phase weights are applied after FFT-based channelization, enabling simultaneous multi-beam operation. The coordination layer computes and distributes beamforming weight vectors at update rates sufficient for electronically steered operation.

Calibration Framework: Array calibration in SPARC is performed in two stages. Factory calibration establishes per-element gain and phase offsets using a known reference signal. Field calibration uses mutual coupling measurements and/or an external calibration source to track drift over time and temperature. Both calibration datasets are stored per-node and applied in real-time by the FPGA processing pipeline.

> Scaling Advantage: Unlike monolithic array processors, SPARC's distributed architecture allows incremental capacity expansion. A 16-element prototype can be field-upgraded to 48 or 96 elements by adding nodes and extending the WR fabric — with no redesign of the core processing pipeline or timing infrastructure.

Data Transport: Inter-node data movement uses the VITA 49 (VRT) packet format over standard Ethernet, ensuring interoperability with existing SDR ecosystems and processing frameworks. Each RFSoC node's 10 GbE or 100 GbE link carries timestamped, channelized IQ data to the aggregation switch. The VITA 49 context packets embed WR-derived timestamps, enabling sample-accurate alignment at the coordination layer without explicit synchronization handshakes.

05 Heritage & Differentiators

Relationship to Project Phoenix / TSS Solutions

Project Phoenix, developed for TSS Solutions, established many of the core signal processing algorithms and beamforming approaches that SPARC inherits. However, SPARC diverges significantly in its hardware platform, timing architecture, and scalability model. The following comparison highlights the key evolutionary steps from Phoenix to SPARC.

COMPARATIVE ANALYSIS

PARAMETER	PHOENIX / TSS	SPARC / RANK4
DIGITIZER	External ADC modules	**Integrated RFSoC ADC**
SYNC METHOD	Centralized clock dist.	**White Rabbit (<1 ns)**
MAX ELEMENTS	16 (practical limit)	**48+ (modular scaling)**
PROCESSING	External FPGA + GPU	**On-chip RFSoC FPGA**
CLOCK JITTER	~500 ps typical	**<100 ps target**
SCALABILITY	Fixed architecture	**Modular / cascadable**
DATA FORMAT	Proprietary	**VITA 49 / VRT standard**
EPOCH REF	GPS only	**PPS server (multi-source)**
COST / ELEM	High (discrete)	**Reduced (integrated)**

Key Differentiators: SPARC's most significant advancement over Phoenix is the elimination of external digitizer hardware. By leveraging the RFSoC's integrated ADC/DAC, SPARC reduces per-element cost, power consumption, and physical footprint while simultaneously improving signal integrity by shortening the analog signal path. The White Rabbit timing infrastructure replaces Phoenix's centralized clock distribution, removing the cable matching requirement that was Phoenix's primary scaling bottleneck.

Additionally, SPARC's adoption of the VITA 49 data transport standard ensures compatibility with a broad ecosystem of SDR processing tools and frameworks. This standards-based approach reduces integration risk and enables Rank4 to leverage existing processing infrastructure at customer sites. The modular node architecture also supports mixed-mode operation, where a single SPARC deployment can simultaneously support multiple waveforms and beam patterns by partitioning node resources across independent processing chains.

SPARC retains Phoenix's proven beamforming algorithms, calibration methodologies, and signal processing chain, ensuring a low-risk technology transition. The core IP — including adaptive nulling, sidelobe control, and wideband true-time-delay processing — has been validated in Phoenix deployments and is being ported directly to the RFSoC FPGA fabric with optimizations for the UltraScale+ architecture.

05 Implementation Roadmap

PHASE 1 // FOUNDATION Q1-Q2

Single-node bring-up on RFSoC 4x2. Validate ADC performance, clock input from WR endpoint, basic channelization in FPGA fabric. Establish VITA 49 data streaming to host. Baseline noise floor and spurious-free dynamic range measurements.

PHASE 2 // SYNCHRONIZATION Q2-Q3

Multi-node synchronization validation. Deploy WR Grand Master + 4-node WR fabric. Characterize inter-node timing accuracy (<1 ns target). Implement PPS-aligned capture triggers. Demonstrate coherent 16-element beamforming with measurable array gain.

PHASE 3 // SCALE Q3-Q4

Expand to 48-element configuration (12 nodes). Validate cascaded WR fabric operation. Implement full calibration framework including mutual coupling compensation. Demonstrate simultaneous multi-beam operation and adaptive nulling across the full aperture.

PHASE 4 // PRODUCTION Q4+

Ruggedization, thermal qualification, and production packaging. Develop deployment automation and node management software. Customer integration testing. Explore 96+ element configurations for advanced applications.

06 Conclusion

Project SPARC represents a strategic investment by Rank4 Signal Systems in the next generation of scalable, coherent phased array technology. By building on the proven signal processing heritage of Project Phoenix and combining it with the integration density of the RFSoC platform and the timing precision of White Rabbit, SPARC delivers a fundamentally more capable and cost-effective architecture for large-aperture array systems.

The distributed, modular nature of SPARC ensures that deployments can scale incrementally from prototype to production, reducing program risk and enabling early capability delivery. Rank4's commitment to open standards (VITA 49, White Rabbit, standard Ethernet) ensures long-term interoperability and positions SPARC as a platform for continuous evolution as array processing requirements grow.

> For Further Information: Contact Rank4 Signal Systems at Rank-4.com for detailed technical specifications, partnership inquiries, or demonstration scheduling.

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PROJECT SPARC

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